Supporting Information

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S.1 Methods

S.1.1 The EAGGL model

EAGGL (Enrichment Analysis Gene Grouping and LLMs) is the latent-structure stage that follows PIGEAN. Whereas PIGEAN estimates the probability that each gene is relevant to a trait and the degree to which each gene annotation predicts trait-relevance, the goal of EAGGL is to decompose those trait-associated gene and annotation patterns into a small number of latent mechanisms.

Let genes be indexed by $i = 1 \ldots N$, gene annotations or gene sets by $j = 1 \ldots M$, phenotypes by $t = 1 \ldots T$, and latent mechanisms by $k = 1 \ldots K$. Let $X_{ij} \geq 0$ denote the value of annotation j for gene i. In the simplest setting, X_{ij} is binary gene-set membership, although weighted annotations are also permitted. Let $p_i^{(u)} \in [0,1]$ denote the gene-level support for gene i for anchor user u, and let $q_j^{(u)} \in [0,1]$ denote the corresponding annotation-level or gene-set-level support for annotation j for anchor user u. The index $u = 1 \ldots U$ is a generic anchor index; depending on the workflow, it may refer to the default phenotype, a supplied positive-control pseudo-phenotype, one or more anchor phenotypes, one or more anchor genes, or an anchor gene set.

The basic biological assumption is that trait-relevant genes and trait-relevant annotations co-occur because they load on a small number of shared mechanisms. In the gene-by-annotation workflow family, we therefore posit nonnegative gene weights W_{ik} and nonnegative annotation weights H_{jk} such that

$$X_{ij} \approx \sum_{k=1}^{K} W_{ik} H_{jk} + \epsilon_{ij}, \tag{S1}$$

where ϵ_{ij} is an error term. A factor k is interpreted as a latent mechanism when a coherent set of genes has large W_{ik} values and a coherent set of annotations has large H_{jk} values.

The entries of X are not equally informative for a given trait. For example, an annotation that is itself strongly implicated by PIGEAN should contribute more strongly to inference than an annotation with no trait support; likewise, genes with high trait support should contribute more than genes with no support. EAGGL therefore fits the factorization under a weighted Gaussian likelihood:

$$\Pr(X \mid W, H, \Omega) \propto \exp\left(-\frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{M} \Omega_{ij} \left(X_{ij} - \sum_{k=1}^{K} W_{ik} H_{jk}\right)^2\right), \tag{S2}$$

where $\Omega_{ij} \geq 0$ is a cell-specific weight. In the simplest single-trait setting, one can think of Ω_{ij} as proportional to $p_i q_j$, where p_i is gene support and q_j is annotation support. More generally, when several anchor users are present, EAGGL uses the additive form

$$\Omega_{ij} = \sum_{u=1}^{U} p_i^{(u)} q_j^{(u)}. \tag{S3}$$

This gives higher weight to entries supported simultaneously by the relevant genes and the relevant annotations for the chosen anchor set.

Although EAGGL is often motivated heuristically by factorizing a matrix of the form $X_{ij}p_iq_j$, the weighted likelihood in Eq. (S2) is the more useful mathematical form. It makes explicit that EAGGL seeks a factorization of the underlying annotation matrix while emphasizing entries most relevant to the trait or anchor set. This also avoids repeatedly materializing a different dense weighted matrix for every anchor user.

S.1.2 Fitting the model

Because the factor matrices are latent, we infer them from the observed nonnegative matrix and the support-derived weights. The factorization is regularized by automatic relevance determination (ARD), which allows the effective number of factors to be learned from the data rather than fixed in advance.

Specifically, for each factor k we introduce a nonnegative relevance scale λ_k and use the prior

$$W_{ik} \mid \lambda_k \sim \mathrm{HalfNormal}\left(0, \lambda_k/\phi\right), \qquad H_{jk} \mid \lambda_k \sim \mathrm{HalfNormal}\left(0, \lambda_k/\phi\right), \tag{S4}$$

together with a factor-specific ARD prior

$$\lambda_k \sim \mathrm{InvGamma}(a_0, b_0). \tag{S5}$$

Here ϕ controls the baseline sparsity of the factor loadings, while a_0 and b_0 control the tendency of entire factors to be shrunk toward low relevance. Small values of λ_k correspond to factors that explain little of the observed matrix after accounting for the regularization. ARD is a soft relevance shrinkage mechanism rather than a guarantee that every low-mass column is driven exactly to zero.

Up to additive constants, maximizing the posterior is equivalent to minimizing

$$\mathcal{L}(W, H, \lambda) = \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{M} \Omega_{ij} \left(X_{ij} - \sum_{k=1}^{K} W_{ik} H_{jk} \right)^2 + \frac{\phi}{2} \sum_{k=1}^{K} \frac{\|W_{\cdot k}\|_2^2 + \|H_{\cdot k}\|_2^2 + 2b_0}{\lambda_k} + C \sum_{k=1}^{K} \log \lambda_k, \tag{S6}$$

where

$$C = \frac{N+M}{2} + a_0 + 1. \tag{S7}$$

Thus, EAGGL uses the same basic intuition as other Bayesian nonnegative matrix factorization procedures with ARD: the reconstruction term encourages faithful explanation of the observed matrix, while the ARD penalty favors retaining only factors whose contribution is large enough to overcome whole-factor shrinkage.

The number of retained mechanisms is therefore not simply the user-specified maximum candidate rank. Rather, EAGGL begins from a candidate rank K and then uses both the posterior relevance scales and post-fit loading mass to distinguish interpretable factors from the low-mass ARD tail. In this way, the effective number of mechanisms is learned adaptively while retaining explicit diagnostics for the fitted tail.

S.1.3 Single-trait gene-by-annotation factorization

The default EAGGL workflow operates on the gene-by-annotation matrix X together with a single trait-specific gene support vector p_i and annotation support vector q_j. In this case $U = 1$ and the weight matrix simplifies to

$$\Omega_{ij} = p_i q_j. \tag{S8}$$

The fitted factor matrices W and H may be interpreted as:

- W_{ik}: the degree to which gene i participates in mechanism k;
- H_{jk}: the degree to which annotation j captures mechanism k.

This is the primary EAGGL model and corresponds to the scientific setting in which one seeks to decompose the trait-associated gene-by-annotation landscape into a small number of mechanisms.

A useful special case is when the input support is not obtained from a standard PIGEAN trait run, but instead from a manually supplied gene list. In that standalone workflow, EAGGL first uses the loaded gene universe from X as the enrichment background and computes a hypergeometric enrichment p-value for every annotation or gene set. If M is the number of genes in the loaded universe, n is the number of supplied genes present in that universe, N_j is the size of annotation j, and k_j is the overlap between annotation j and the supplied gene list, then

$$p_j^{(\text{hyper})} = \Pr(\text{Hypergeom}(M, n, N_j) \geq k_j)\,. \tag{S9}$$

After Benjamini–Hochberg correction, annotations with $q_j \leq q_{\max}$ are retained. Their annotation-level support is then defined by

$$q_j^{(\text{standalone})} = \frac{-\log p_j^{(\text{hyper})}}{\sqrt{N_j}}, \tag{S10}$$

while the gene-level support vector is binary:

$$p_i^{(\text{standalone})} = \begin{cases} 1 & \text{if gene } i \text{ is in the supplied list,} \\ 0 & \text{otherwise.} \end{cases}$$

All genes appearing in any retained annotation are then kept in the final factoring matrix. The factorization itself is unchanged; only the way the gene-level and annotation-level support vectors are constructed differs from the standard PIGEAN-driven path.

S.1.4 Phenotype-anchored extensions of the gene-by-annotation model

In some applications, one wishes to condition factorization on several anchor phenotypes rather than on a single default trait. Let P_{it} denote a nonnegative gene-by-phenotype support matrix and Q_{jt} denote a nonnegative annotation-by-phenotype support matrix derived from phenotype-resolved statistics. If the anchor phenotype set is $A \subseteq \{1, \dots, T\}$, then each $t \in A$ defines an anchor user, with

$$p_i^{(u)} = P_{it}, \qquad q_j^{(u)} = Q_{jt}.$$

The weight matrix again takes the additive form of Eq. (S3):

$$\Omega_{ij}^{(A)} = \sum_{t \in A} P_{it} Q_{jt}. \tag{S11}$$

The factorization itself remains the same as Eq. (S1). The difference is that the model is now conditioned on a selected phenotype set rather than on a single default trait.

When the goal is to summarize relevance to any phenotype in a large panel rather than to a manually specified subset, the anchor set is first collapsed to a single aggregate user and factorization proceeds under the same weighted model. Thus, the difference between multi-phenotype anchoring and any-phenotype anchoring lies in whether the phenotype-resolved support vectors are retained separately or combined before fitting; the underlying factor model is unchanged.

S.1.5 Gene-anchored and gene-set-anchored gene-set by phenotype factorization

The second major EAGGL workflow family factors a nonnegative gene-set-by-phenotype matrix rather than the original gene-by-annotation matrix. Let $Y_{jt} \geq 0$ denote a matrix whose rows are gene sets and whose columns are phenotypes, typically derived from phenotype-resolved PIGEAN statistics. EAGGL then posits

$$Y_{jt} \approx \sum_{k=1}^{K} H_{jk} Z_{tk} + \epsilon_{jt}, \tag{S12}$$

where H_{jk} is again the loading of annotation j on factor k, while Z_{tk} is now the phenotype loading of factor k.

As above, the factorization is fitted under a weighted Gaussian likelihood. Let $q_j^{(u)}$ denote gene-set support and let $r_t^{(u)}$ denote phenotype support induced by anchor user u. Then

$$\Omega_{jt} = \sum_{u=1}^{U} q_j^{(u)} r_t^{(u)}. \tag{S13}$$

The anchor users are defined differently for the different workflows in this family:

- In single-gene anchoring and multi-gene anchoring, u indexes one or more selected anchor genes.
- In any-gene anchoring, the selected anchor genes are first collapsed into a single aggregate user.
- In gene-set anchoring, the user is a supplied anchor gene set that induces phenotype support through the gene-PheWAS statistics.

Thus, the gene-anchored workflows and the phenotype-anchored workflows are not different factorization algorithms. They are different constructions of the matrix and weights supplied to the same ARD-regularized nonnegative factor model.

S.1.6 Implicit factor relevance and any-anchor summaries

After fitting, EAGGL can compute factor-level relevance scores that summarize how strongly each factor is supported by the selected trait or anchor set. These are downstream summaries derived from the fitted factor basis and are not themselves part of the factorization objective. In the default `--gene-stats-in` / `--gene-set-stats-in` workflow, the input statistics define an implicit anchor named `input_gene_stats`; no separate anchor flag is required. The public factor output no longer reports the legacy `any_relevance` alias; anchor summaries are exposed through the canonical anchor_any_joint and anchor_any_marginal columns only when both joint and marginal anchor linkage summaries are available.

For the gene-by-annotation workflow family, let $W \in \mathbb{R}_+^{N\times K}$ denote the fitted gene-by-factor matrix, and let $p^{(u)} \in [0,1]^N$ denote the gene-level support vector for anchor user u. The anchor-specific relevance of the factors to user u is defined by bounded nonnegative regression:

$$r^{(u)} = \arg\min_{0\le r\le 1} \left\| p^{(u)} - Wr \right\|_2^2 . \tag{S14}$$

For the gene-set-by-phenotype workflow family, let $Z \in \mathbb{R}_+^{T\times K}$ denote the fitted phenotype-by-factor matrix, and let $p^{(u)} \in [0,1]^T$ denote the phenotype support vector for anchor user u. The factor relevance is then defined analogously:

$$r^{(u)} = \arg\min_{0\le r\le 1} \left\| p^{(u)} - Zr \right\|_2^2 . \tag{S15}$$

When more than one anchor user is present, EAGGL summarizes the bounded projection relevance of factor k to any selected anchor by a noisy-OR-style aggregation:

$$r_k^{\text{any}} = 1 - \prod_{u=1}^{U} \left(1 - r_k^{(u)}\right) . \tag{S16}$$

This has the intended semantics that r_k^{any} is a bounded coefficient summary of how strongly factor k links to at least one selected anchor under the relevance model above. In the single-user case, Eq. (S16) reduces to the anchor-specific relevance itself.

When anchor phenotypes are present and canonical trait linkage is also computed, EAGGL reports anchor_any_joint and anchor_any_marginal summaries in factors.out(.gz) derived from the canonical linkage calculations in Section S.2. If only the older joint-style relevance calculation is available, these anchor-any columns are not emitted rather than aliasing marginal to joint. The per-anchor factors_anchor.out(.gz) table reports the corresponding joint and marginal values for each anchor separately, using NA for marginal when no marginal summary was computed.

S.2 Trait linkage and factor-specific phenotype enrichment

After fitting the main factorization, EAGGL can relate latent factors to additional human phenotypes in two conceptually distinct ways. The primary public annotation layer is a descriptive *trait linkage* step, which estimates how strongly each phenotype's thresholded high-confidence support profile is represented by each factor. The secondary expert-only layer is an inferential *enrichment* step, which asks whether a phenotype is more concentrated in one factor than expected after accounting for the anchor trait and optional covariates. These two quantities answer different scientific questions and are therefore not expected to coincide exactly.

S.2.1 Canonical trait linkage by support-normalized thresholded projection

The external gene-by-phenotype support table can be very large. In practice, EAGGL therefore consumes a sparse thresholded representation that retains only high-confidence entries. The phenotype profile used for post-factor annotation is therefore not the full unthresholded phenotype surface. It is the thresholded support profile on a single chosen support surface. Publicly, EAGGL defaults to `--trait-linkage-source combined` and `--trait-linkage-threshold 1.0`, using a strict threshold rule (`source value` $> \theta_{\text{trait}}$). Expert overrides can choose `log_bf`, `prior`, or `auto` fallback resolution.

Let b denote a feature basis shared by the factors and a trait-specific support vector. In the public workflow, b is the gene basis whenever gene-factor loadings are available. EAGGL uses the gene-set basis only as an expert or fallback mode when the gene basis is unavailable or the user explicitly requests `--project-phenos-from-gene-sets`. Let

$$W^{(b)} \in \mathbb{R}_{+}^{F_b \times K}$$

denote the raw factor-loading matrix on that feature space, where F_b is the number of features in basis b, and let

$$s_t^{(b)} \in \mathbb{R}_{+}^{F_b}$$

denote the raw thresholded trait-support profile for phenotype t on that same basis. These raw trait and factor vectors are not interpreted as biological probability distributions and are not required to sum to 1. Their total support and total mass are meaningful quantities that are preserved separately from the copied normalized vectors used for trait-factor matching.

By default, EAGGL uses a *weighted thresholded* profile: retained source-support values are used for entries that strictly exceed the selected threshold and zero otherwise. As an expert sensitivity analysis, one may instead use a *binary thresholded* profile, in which retained entries are coded as 1 and absent entries as 0 before normalization.

For linkage, EAGGL first computes the total thresholded trait support on the full thresholded support surface, before applying the factor mask. Define the total thresholded trait support

$$A_t^{(b)} = \sum_{f=1}^{F_b^{\text{full}}} s_{tf}^{(b,\text{full})}, \tag{S17}$$

where $s_t^{(b,\text{full})}$ denotes the full thresholded support surface on basis b before masking. Let $\mathcal{M}_b$ denote the retained factor mask on that basis, and let

$$s_{t,\mathcal{M}}^{(b)}$$

denote the masked thresholded support profile restricted to the retained factor basis rows. EAGGL then builds a full-space matching target by zeroing support outside $\mathcal{M}_b$, and keeps the normalization denominator as the full thresholded trait strength. Thus the normalized masked trait profile is

$$q_t^{(b)} = \frac{s_{t,\mathcal{M}}^{(b)}}{A_t^{(b)} + \epsilon}, \tag{S18}$$

where $\epsilon > 0$ is a small constant to avoid division by zero. After masking, $\sum_f q_{tf}^{(b)}$ generally equals the retained support fraction, not 1:

$$\sum_f q_{tf}^{(b)} = \frac{\sum_f s_{t,\mathcal{M}f}^{(b)}}{A_t^{(b)} + \epsilon}.$$

Likewise, each factor basis vector is copied into an internally normalized profile,

$$S_k^{(b)} = \sum_{f=1}^{F_b} W_{fk}^{(b)}, \tag{S19}$$

$$b_{\bullet k}^{(b)} = \frac{W_{\bullet k}^{(b)}}{S_k^{(b)} + \epsilon}. \tag{S20}$$

The raw factor loadings $W^{(b)}$ and their totals $S_k^{(b)}$ are preserved for reporting; only the copied profiles $b_{\bullet k}^{(b)}$ are used for matching.

EAGGL then reports two linkage summaries from the same internal matching inputs. The *marginal coefficient* of factor k for trait t is the one-factor bounded projection

$$m_{tk}^{(b)} = \arg\min_{0\le c\le 1} \left\| q_t^{(b)} - b_{\bullet k}^{(b)} c \right\|_2^2, \tag{S21}$$

and the marginal overlap companion is the direct shape overlap

$$o_{tk}^{(b)} = \left(q_t^{(b)}\right)^\top b_{\bullet k}^{(b)}. \tag{S22}$$

and the *joint coefficient* is the all-factor competitive projection

$$j_t^{(b)} = \arg\min_{c\ge 0,\ \mathbf{1}^\top c\le 1} \left\| q_t^{(b)} - b^{(b)} c \right\|_2^2. \tag{S23}$$

We also report the uncaptured normalized mass

$$\rho_t^{(b)} = \max\left(0,\ 1 - \sum_{k=1}^{K} j_{tk}^{(b)}\right). \tag{S24}$$

The coefficient $m_{tk}^{(b)}$ is interpreted as a standalone support-normalized projection coefficient linking phenotype t to factor k when that factor is viewed alone. Because it is divided by the factor self-norm, it remains coefficient-like and can be sensitive to factor breadth. The overlap $o_{tk}^{(b)}$ is a direct shape-overlap metric and is less affected by broad factor profiles. The coefficient $j_{tk}^{(b)}$ is interpreted as the corresponding competitive projection coefficient for factor k after the other factors compete. These coefficients are not exact overlap or captured-support masses. In contrast to a projection normalized only on the retained masked slice, Eqs. (S21) and (S23) preserve the scale of the full thresholded trait while still solving on the retained factor basis. Broad phenotypes can therefore have large total thresholded support $A_t^{(b)}$ without automatically receiving inflated linkage coefficients when only a small masked subset of their support survives into the factor basis.

For reporting, EAGGL writes:

- `trait_total_support` $= A_t^{(b)}$
- `retained_trait_support` $= \sum_f s_{t,\mathcal{M}f}^{(b)}$
- `retained_fraction` = `retained_trait_support` / `trait_total_support`
- `trait_n_eff` $= \left(\sum_f s_t^{(b)}\right)^2 / \sum_f \left(s_t^{(b)}\right)^2$
- `retained_n_eff` $= \left(\sum_f s_{t,\mathcal{M}f}^{(b)}\right)^2 / \sum_f \left(s_{t,\mathcal{M}f}^{(b)}\right)^2$
- `joint_coefficient` $= j_{tk}^{(b)}$

- `marginal_coefficient` $= m_{tk}^{(b)}$
- `marginal_overlap` $= o_{tk}^{(b)}$
- `joint_coefficient_support_mass` = `trait_total_support` × `joint_coefficient`, a coefficient-scaled support total rather than an exact captured-support mass
- `marginal_coefficient_support_mass` = `trait_total_support` × `marginal_coefficient`, a coefficient-scaled support total rather than an exact captured-support mass
- `marginal_overlap_support_mass` = `trait_total_support` × `marginal_overlap`, an overlap-scaled support total
- `factor_tier` and `combined_mass_fraction` report the post-fit interpretation tier and global factor mass fraction.
- `factor_total_mass` $= S_k^{(b)}$ in the factor summary output
- `factor_n_eff` $= \left(\sum_f W_{fk}^{(b)}\right)^2 / \sum_f \left(W_{fk}^{(b)}\right)^2$
- `factor_top_share` $= \max_f W_{fk}^{(b)} / S_k^{(b)}$
- `factor_top10_share` = top-10 loading mass divided by $S_k^{(b)}$
- `broad_factor_flag` = 1 when `factor_n_eff` ≥ 500 and `factor_top_share` ≤ 0.01

The effective-size diagnostics complement the raw thresholded feature counts. A phenotype can have many retained nonzero genes but still have `trait_n_eff` or `retained_n_eff` close to 1 when most of its support is concentrated on one or a few genes. These quantities therefore expose concentration of support mass rather than only thresholded breadth.

Interpretation. Canonical trait linkage is a descriptive quantity. The marginal coefficients answer:

> "How much does factor k link to phenotype t when factor k is viewed alone?"

The joint coefficients answer:

> "How is the retained high-confidence support profile of phenotype t represented after the factors compete as a mixture of latent mechanisms?"

These are not p-values, and they should not be interpreted as calibrated posterior probabilities that phenotype t is formally associated with factor k.

EAGGL also reports retained-basis diagnostics alongside the linkage scores: total thresholded trait support, retained masked trait support, retained fraction, total thresholded feature count, retained masked feature count, support effective sizes, and a low-retention flag for traits with very small or highly concentrated retained support.

S.2.2 Choice of projection basis across workflow families

For the default gene $\times$ gene-set workflow family (F1–F5), the preferred basis for canonical trait linkage is the gene basis. In that case, $W^{(b)}$ is the gene-by-factor loading matrix and $s_t^{(b)}$ is the thresholded phenotype-specific gene-support vector. If gene-level support is unavailable or gene-level factor loadings are absent, EAGGL falls back to the gene-set basis, in which $W^{(b)}$ is the gene-set-by-factor loading matrix and $s_t^{(b)}$ is the thresholded phenotype-specific gene-set support vector.

For the gene-set $\times$ phenotype workflow family (F6–F9), public trait linkage still uses the same projection definition whenever an external trait profile is provided, typically on the gene-set basis. Thus, the canonical linkage definition is common across workflows: EAGGL always projects one thresholded support profile onto a factor basis defined on one shared feature space, then reports both marginal and joint coefficients from those same internal matching inputs.

S.2.3 Factor-specific phenotype enrichment with thresholded binary outcomes

A second, distinct summary asks whether phenotype t is selectively enriched in genes belonging to factor k, after accounting for the main anchor trait and optional covariates. Because the standard phenotype input is thresholded, the default factor-specific enrichment analysis is based on a binary thresholded outcome rather than on truncated continuous support values. Let

$$h_{it} = \mathbb{I}\{\text{combined phenotype support for gene } i \text{ and phenotype } t \text{ exceeds } c_0\}$$

denote the resulting thresholded phenotype-hit indicator. Let W_{ik} denote the gene-membership weight of factor k for gene i, let a_i^{direct} denote the direct support of the main anchor trait for gene i, and let C_i denote optional gene-level covariates. The default model fits, for each phenotype t and factor k,

$$h_{it} = \alpha_{tk} + \beta_{tk} W_{ik} + \gamma_t a_i^{\text{direct}} + \eta_t^{\top} C_i + \varepsilon_{it}. \tag{S25}$$

This regression is fit one factor at a time, i.e. as a marginal anchor-adjusted binary model. The coefficient β_{tk} is interpreted as the factor-specific enrichment of phenotype t beyond what is already captured by the anchor trait's direct support.

Interpretation. Unlike the canonical linkage weights in Eqs. (S21) and (S23), the coefficients in Eq. (S25) are incremental effects. They answer the question:

> "Is phenotype t enriched in factor k beyond the anchor trait's direct gene support?"

This is an inferential or hypothesis-testing quantity rather than a descriptive loading.

Recommended default and scope. The binary thresholded model is the default because it is defensible under sparse thresholded phenotype input and easier to interpret than a joint all-factor fit. EAGGL reports robust uncertainty by default, typically using HC3-type standard errors or chromosome-clustered uncertainty when location information is available. The resulting p-values are still approximate, because the thresholded phenotype-hit process is derived from a combined-support quantity that already reflects gene-set-mediated indirect support.

An expert may instead fit: (i) a marginal binary model without an anchor covariate; (ii) a joint all-factor binary model with direct anchor adjustment; or (iii) legacy continuous direct/combined regressions. Using combined anchor support as the covariate is also an expert-only approximation, because then both the outcome and the covariate inherit related gene-set-mediated structure.

When multiple factor-specific enrichment models are requested in one run, EAGGL appends them into a single output table and tags each row with the model name, factor-model scope, outcome surface, and anchor-adjustment setting so that marginal, joint, and legacy analyses are not conflated downstream.

S.2.4 Relationship between canonical trait linkage and factor-specific enrichment

The two phenotype summaries serve complementary roles:

- **Canonical trait linkage** gives bounded marginal and joint support-normalized projection coefficients describing how strongly the retained high-confidence phenotype profile links to each factor. This is the primary annotation layer for interpreting factors with external phenotypes.
- **Factor-specific enrichment regression** tests whether that phenotype profile is unusually concentrated in one factor beyond what is already captured by the anchor trait and optional covariates. This is a secondary, expert-oriented summary.

Consequently, the main phenotype labels and top-phenotype annotations for a factor should be based on canonical trait-linkage weights, whereas regression coefficients and associated significance measures should be interpreted as downstream enrichment diagnostics rather than as the primary phenotype annotation of the factor.

S.2.5 Repeated restarts, consensus factorization, and structural selection of ϕ

The EAGGL objective is non-convex, and therefore different random initializations can converge to different local optima. For this reason, EAGGL is typically fit under repeated random restarts. The simplest use of repeated restarts is to retain the fitted solution with the best objective value. A more stringent alternative is consensus factorization, in which factors are first matched across runs and then aggregated only if they are reproduced consistently across restarts. Let $H^{(r)}$ denote the annotation loading matrix from restart r after ARD thresholding. In consensus mode, EAGGL normalizes the factor columns, aligns factors across restarts by maximum total similarity, and aggregates matched factors only when they appear in a sufficient fraction of runs.

The hyperparameter ϕ also affects the effective number and sparsity of retained factors, even though ARD is responsible for pruning irrelevant components. EAGGL therefore treats ϕ as a structural model-selection parameter. In the current public automatic tuning mode, users specify a target median effective gene support for primary factors, and EAGGL searches for a ϕ whose fitted mechanisms have that approximate gene-module size.

Let R be the number of restarts, let $K^{(r)}$ be the retained factor count in restart r, and let K_{mode} be the modal retained factor count. Let

$$\mathcal{R}_{\text{mode}} = \{r : K^{(r)} = K_{\text{mode}}\},$$

and let $m = |\mathcal{R}_{\text{mode}}|$. The restart-support statistic is

$$s(\phi) = \frac{m}{R}. \tag{S26}$$

When $m \geq 2$, factors are aligned across modal restarts by maximum total cosine similarity and restart stability is defined as the mean matched-factor cosine across the modal restart set. When $m < 2$, stability is undefined and the candidate is not considered restart-stable.

Within each modal restart, EAGGL measures redundancy on the gene loading matrix when available, with fallback to the gene-set and phenotype loading matrices. These phi-selection metrics are computed over the primary factor set by default so that low-mass ARD-tail columns do not dominate model selection; an audit option can instead score all fitted columns. After clipping negligible weights and renormalizing each factor loading vector, overlap between vectors u and v is measured by weighted Jaccard:

$$J_w(u, v) = \frac{\sum_\ell \min(u_\ell, v_\ell)}{\sum_\ell \max(u_\ell, v_\ell)}. \tag{S27}$$

For metric-scope factor k, let $d_k = \max_{\ell \neq k} J_w(u_k, u_\ell)$. EAGGL records

$$d_{\max} = \max_k d_k, \qquad d_{0.9} = Q_{0.9}(d_1, \ldots, d_K), \tag{S28}$$

where $Q_{0.9}$ denotes the 90th percentile.

For each candidate, EAGGL also summarizes factor mass. If m_k denotes the total nonnegative loading mass carried by factor k, then

$$K_{\text{mass}} = \frac{\left(\sum_k m_k\right)^2}{\sum_k m_k^2} \tag{S29}$$

is reported as an effective factor count. Primary factors are those whose mass fraction exceeds the configured primary-factor floor. For primary factor k, let normalized gene weights be $\tilde{w}_{ik} = W_{ik}/\sum_i W_{ik}$. Its effective gene support is

$$S_k = \frac{1}{\sum_i \tilde{w}_{ik}^2}. \tag{S30}$$

The candidate-level size statistic is $S_\phi = \text{median}_{k \in \mathcal{P}_\phi} S_k$, where $\mathcal{P}_\phi$ is the set of primary factors. Given user target $S^\star$, EAGGL scores size mismatch on log scale,

$$E_\phi = \left|\log S_\phi - \log S^\star\right|. \tag{S31}$$

Candidates must be non-collapsed, have enough primary factors, satisfy restart and redundancy constraints, and pass the fit-loss guardrail. Restart stability is also matched over the same metric factor scope, which is primary by default and independent of the output-printing scope. Among acceptable uncapped candidates with defined S_ϕ, EAGGL first restricts to candidates within the configured size tolerance when any exist, then chooses the largest ϕ in that tolerance set. If none are in tolerance, it chooses the candidate with smallest E_ϕ, using larger ϕ, lower tail fraction, lower filtered fraction, lower redundancy, better fit, and lower primary-factor spike diagnostics as tie-breakers. Candidate values are explored around the initial ϕ by target-aware bracketing: if fitted factors are too small, larger ϕ values are proposed; if fitted factors are too large, smaller ϕ values are proposed; once the target is bracketed, geometric midpoints refine the search.

The selected ϕ, candidate diagnostics, search thresholds, primary-factor gene-support summaries, redundancy summaries, factor-mass summaries, and convergence diagnostics are written to the run log and parameter output. Per-factor diagnostics for each tested ϕ can also be written for audit and review.

EAGGL separates fitted factors from reported factors. Let m_k denote the combined nonnegative gene and annotation loading mass for factor k, and let $\pi_k = m_k/\sum_\ell m_\ell$. A factor is classified as primary when $\pi_k \geq 0.005$, secondary when $0.001 \leq \pi_k < 0.005$, and filtered otherwise. The default user-facing factor and cluster outputs print only primary factors, because low-mass secondary and filtered factors usually represent the ARD tail rather than interpretable mechanisms. This reporting scope is controlled separately from the phi-selection metric scope. The exhaustive factor-metrics outputs remain available for auditing all fitted columns. The `factors.out(.gz)` table records both `factor_tier` and `combined_mass_fraction` next to `lambda`; widening the output scope can include secondary or all factors without changing the fitted model or renumbering factor identifiers.

S.2.6 Optimizations for scaling EAGGL to large phenotype panels

Although the model is defined in terms of weighted nonnegative factorization, a direct implementation that explicitly constructs separate dense matrices for every anchor user would be inefficient. EAGGL therefore uses the weighted objective of Eq. (S2) directly, rather than materializing large user-specific tensors. This preserves the model while substantially reducing memory use.

Similarly, phenotype panels and annotation collections can be large enough that a naive factorization would be dominated by nearly null rows or by many near-duplicate annotations. EAGGL therefore distinguishes three annotation sets during factor fitting: retained annotations, discovery annotations, and projected annotations. Retained annotations are the rows that survive ordinary hard support and quality filters. Discovery annotations are a redundancy-balanced subset of family leaders used to learn the shared factor basis W. Projected annotations are the full retained set, obtained afterward by fixed-W nonnegative projection.

In the default discovery mode, retained annotations are processed in support-ranked order and greedily assigned to similarity-defined families. Let F_c denote discovery family c, let r_c denote its leader, and let $s(j, r_c) \in [0, 1]$ be the same clipped annotation similarity used for assignment. A candidate annotation starts a new family when its maximum similarity to existing leaders is less than the discovery similarity threshold ρ_{disc}; otherwise it is assigned to the nearest leader. For singleton families $m_{c,\text{eff}} = 1$. For larger families,

$$\bar{s}_c = \frac{1}{|F_c| - 1} \sum_{j \in F_c \setminus \{r_c\}} s(j, r_c), \qquad m_{c,\text{eff}} = \frac{|F_c|}{1 + (|F_c| - 1)\bar{s}_c}. \tag{S32}$$

The default effective-size discovery support for anchor user u is

$$\tilde{q}_c^{(u)} = q_{r_c}^{(u)} m_{c,\text{eff}}. \tag{S33}$$

Thus exact duplicates contribute approximately one representative unit, whereas weakly redundant families can contribute closer to their family size. The conservative `log_effective_size` mode uses $q_{r_c}^{(u)}(1 + \log m_{c,\text{eff}})$, and `none` uses the representative support $q_{r_c}^{(u)}$ without a family-size multiplier. EAGGL records `in_discovery`, `discovery_family_size`, `discovery_family_mean_similarity`, `discovery_family_effective_size`, and `discovery_weight` in cluster outputs so the discovery objective can be audited directly.

Because discovery support weights change the data term scale, the implementation computes internal ARD scaling quantities from the row-weighted discovery matrix rather than the raw discovery matrix. Equivalently, if V is the discovery matrix and D is the diagonal matrix of discovery row weights, scale-dependent quantities use $D^{1/2}V$. This keeps the effective ϕ scaling, ARD prior floor, and related heuristics aligned with the weighted objective. Finally, output cluster tables first apply the requested factor output scope and then omit gene and gene-set rows whose maximum raw factor loading among the printed factors is below the configurable `--cluster-row-min-max-loading` threshold; these are reporting filters and do not alter the fitted factor matrices.

S.2.7 Interpretation and labeling of latent mechanisms

The outputs of EAGGL are the fitted factor loadings over genes, annotations, and, when applicable, phenotypes. These loadings constitute the statistical model output. Human-readable labels for factors are downstream summaries rather than part of the factorization model itself.

If $\mathcal{S}_k$, $\mathcal{G}_k$, and $\mathcal{P}_k$ denote the highest-weight annotations, genes, and phenotypes for factor k, then a label may be viewed abstractly as a post hoc summary function

$$L_k = \mathcal{L}(\mathcal{S}_k, \mathcal{G}_k, \mathcal{P}_k).$$

Such labels may be generated deterministically from the top terms or by a separate language-model-based summarization step. In either case, the labeling procedure is downstream of the core statistical model and should be interpreted as an aid to biological interpretation rather than as part of the inference procedure itself.